Topology of Neural-Network Layers for Out-of-Distribution Detection

Implementation and evaluation of six topological approaches on one shared model · Dat Bui · October 2026 · code: github.com/Datquangbui1011/Research

1. Goal and shared model

The goal was to find a model used in the six papers of the literature report, implement every approach on it, and evaluate each approach as an out-of-distribution (OOD) detector. Papers [1]–[3] all study MLPs on MNIST, and the pTU paper [3] gives the exact setup. So the shared model is a **Bayesian MLP 784→64→16→10** (ReLU) trained with Bayes-by-Backprop on MNIST, with [3]'s scale-mixture prior ($\sigma_1 = 1$, $\sigma_2 = 0.007$) and early stopping (patience 25). Test accuracy is **97.21 ± 0.09 %** over 3 seeds ([3] reports 97.39 %). The KL term was scaled by 0.1; with the full KL the model stalls at 94.5 %, and [3] does not state its weighting. Deterministic methods use the posterior-mean weights; pTU uses 30 posterior samples.

2. Approaches as implemented

Paper	Object built from the network	OOD score used here
[1] Neural Persistence (Rieck 2019)	Bipartite layer graph of |W| (weights only); 0-dim superlevel PD = MST	Original NP is input-independent (AUROC = 50 by construction). Adaptation: NP of each input's activation graph, z-scored per predicted class
[2] Topological Uncertainty (Lacombe 2021)	Activation graph |W(j,i)·x(i)| per layer; PD = MST weights	Mean over layers of the 2-Wasserstein distance to the class-mean PD of training data
[3] pTU (Yeh & Yang 2025)	Activation-graph PDs under 30 posterior weight samples	Fréchet variance of the 30 PDs, averaged over layers; plus the paper's permutation test (Alg. 2)
[4] Persistent Laplacian (Wolf 2025; PETLS)	In dim 0 with all vertices born first, $\Delta_0^{K_s,K_t}$ = graph Laplacian of K_t	β_0 (zero eigenvalues) and nonzero-spectrum features at 3 thresholds, z-scored per predicted class
[5] Data manifold (Magai & Ayzenberg 2022)	Point cloud of layer activations (input, 2 hidden, logits); Rips PH	Batch: H0/H1 total persistence + TwoNN intrinsic dimension. Per image (adaptation): change in H0/H1 of the 20-NN cloud when x is added
[6] CNN activations (Purvine 2023)	Same point clouds (one vector per image already in an MLP); Rips PDs	Batch: sliced-Wasserstein distance to reference PDs; and the layer-to-layer SW profile

3. Evaluation protocol

In-distribution: 2000 MNIST test images; reference statistics from MNIST training images. OOD sets (2000 each, grayscale 28×28 in [0, 1]): Fashion-MNIST, KMNIST, CIFAR-10, SVHN, Gaussian and Uniform noise (as in [2]), and MNIST-flip (100 pixels inverted, the corruption shift of [2]). Metrics: AUROC (OOD positive) and FPR95 (% of OOD accepted when 95 % of ID is accepted). Baselines: max-softmax confidence (MSP, the baseline of [2]), Bayesian mutual information, and k-NN distance in activation space. Every number is the mean ± std over 3 independently trained models; each seed's metric is first averaged over the 7 OOD sets. † = first 1000 images per set.

4. Results

Per-image detection	AUROC ↑	FPR95 ↓	Batch level (m = images per batch)	m = 10	m = 20
MSP (baseline)	74.1 ± 4.8	74.1 ± 3.0	MSP, batch mean	85.6 ± 6.7	89.0 ± 8.3
Bayesian mutual information	81.4 ± 4.8	60.2 ± 9.0	[2] TU, batch mean	**94.6 ± 0.7**	96.8 ± 0.5
k-NN distance †	88.4 ± 0.9	46.4 ± 1.9	[5] H0/H1 persistence + intrinsic dim.	94.2 ± 0.5	98.9 ± 0.0
[1] NP per input (adaptation)	55.4 ± 5.0	89.2 ± 4.9	[6] SW distance to reference PDs	92.1 ± 0.4	**99.0 ± 0.2**
[2] TU (paper)	79.1 ± 0.8	73.1 ± 3.1	[6] layer-to-layer SW profile	90.2 ± 0.9	94.6 ± 1.6
[2] TU, layers rescaled	87.4 ± 0.6	60.1 ± 1.6			
[3] pTU (paper)	83.0 ± 5.2	39.4 ± 7.0	**[3] pTU permutation test**	**m = 20**	**m = 50**
[4] full persistent Laplacian †	82.6 ± 0.8	59.3 ± 1.5	OOD sets flagged (α = 0.05)	66–100 %	96–100 %
[4] β_0 only (PH part) †	85.8 ± 0.1	50.3 ± 1.6	MNIST test flagged (false alarm)	5 ± 2 %	3 ± 2 %
[5] local ΔH0 (adaptation) †	**96.5 ± 0.4**	**16.9 ± 2.3**			

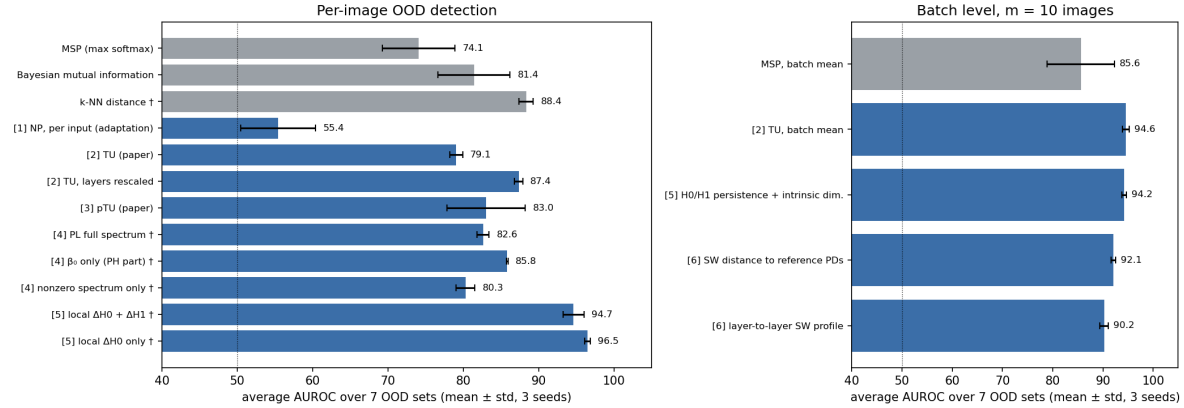


Figure 1. Average AUROC over the 7 OOD sets (mean ± std, 3 seeds). Grey: baselines; blue: topological approaches.

Per-image AUROC	Fashion-MNIST	KMNIST	CIFAR-10	SVHN	Gaussian	Uniform	MNIST-flip
MSP	68.1 ± 4.4	84.1 ± 0.3	68.8 ± 6.2	64.9 ± 11.0	92.7 ± 0.7	59.8 ± 13.0	80.2 ± 1.2
[2] TU	82.2 ± 1.4	71.2 ± 1.7	80.6 ± 1.4	82.2 ± 0.7	95.3 ± 2.0	79.5 ± 7.9	62.6 ± 1.3
[3] pTU	82.1 ± 4.1	82.7 ± 4.9	94.5 ± 2.3	80.7 ± 4.8	46.9 ± 18.2	99.9 ± 0.1	94.5 ± 2.1
[4] β_0 †	84.6 ± 1.3	78.2 ± 1.7	77.8 ± 2.0	84.8 ± 2.3	98.2 ± 1.7	96.6 ± 0.5	80.7 ± 2.1
[5] local ΔH0 †	93.8 ± 0.8	95.3 ± 0.3	99.2 ± 0.1	94.5 ± 1.0	93.5 ± 1.0	100.0 ± 0.0	99.3 ± 0.1

5. Findings

• **Topology beats confidence where the network is confidently wrong.** SVHN, CIFAR-10 and uniform noise receive softmax confidence of about 0.94–1.0, so MSP drops to 60–69 AUROC, while TU, pTU and β_0 reach 78–100. Every approach except [1] beats MSP on average, and the topological scores are also far more stable across training runs (MSP ± 4.8 vs ± 0.1–1.4 for most).

• **[1] Neural persistence cannot detect OOD inputs.** It summarizes the trained weights, not an input. Applied per input it stays near chance (55.4).

• **[2] TU reproduces its paper's claim but is limited by how layers are combined.** The first layer alone reaches about 90 AUROC; averaging raw distances over layers lets large-scale layers dominate. Rescaling each layer by its training mean lifts TU from 79.1 to 87.4 on every seed.

• **[3] pTU is the best paper-defined per-image score (83.0) and its permutation test works as intended:** OOD sets are flagged in 96–100 % of repeats with 50 images, at the nominal 3–5 % false-alarm rate. Weakness: pTU can move in either direction. Gaussian noise often gets lower pTU than MNIST (46.9 ± 18.2), so a one-sided threshold misses it.

• **[4] The persistent Laplacian's extra eigenvalues do not help.** On these triangle-free layer graphs the 0-dim persistent Laplacian is a graph Laplacian. Its harmonic part β_0 alone (85.8) beats the full spectrum (82.6) and TU (79.1); the nonzero eigenvalues take a few discrete values on small unweighted graphs and add noise.

• **[5]/[6] Point-cloud topology is the strongest signal.** At batch level, H0/H1 persistence plus intrinsic dimension ([5]) and SW distances between diagrams ([6]) reach 99 AUROC with 20 images; TU's batch mean ties at 10 images. MNIST's intrinsic dimension shrinks layer by layer (10.6 → 4.6), as [5] reports. Per image, the local ΔH0 score is best overall (96.5 ± 0.4), but it is essentially a refined nearest-neighbour distance across layers: higher-order (H1) features alone are below chance, and [6]'s layer-to-layer profile is weaker than the plain distance to reference diagrams.

Limitations. One small MLP on MNIST (the setting of [1]–[3], not the CNNs of [5]–[6]); 3 seeds; KL term scaled by 0.1; per-image versions of [1] and [5], the MLP version of [6], rescaled TU and β_0-only are my adaptations. Next steps: CNN backbones (CIFAR-10), layer-weighted TU/pTU, and two-sided pTU scores.

[1] Rieck et al., ICLR 2019. [2] Lacombe et al., IJCAI 2021. [3] Yeh & Yang, arXiv:2511.18813. [4] Wolf et al., arXiv:2509.20220; Jones & Wei (PETLS); Botnan & Dong, arXiv:2408.16741. [5] Magai & Ayzenberg, arXiv:2204.08624. [6] Purvine et al., AAAI 2023.